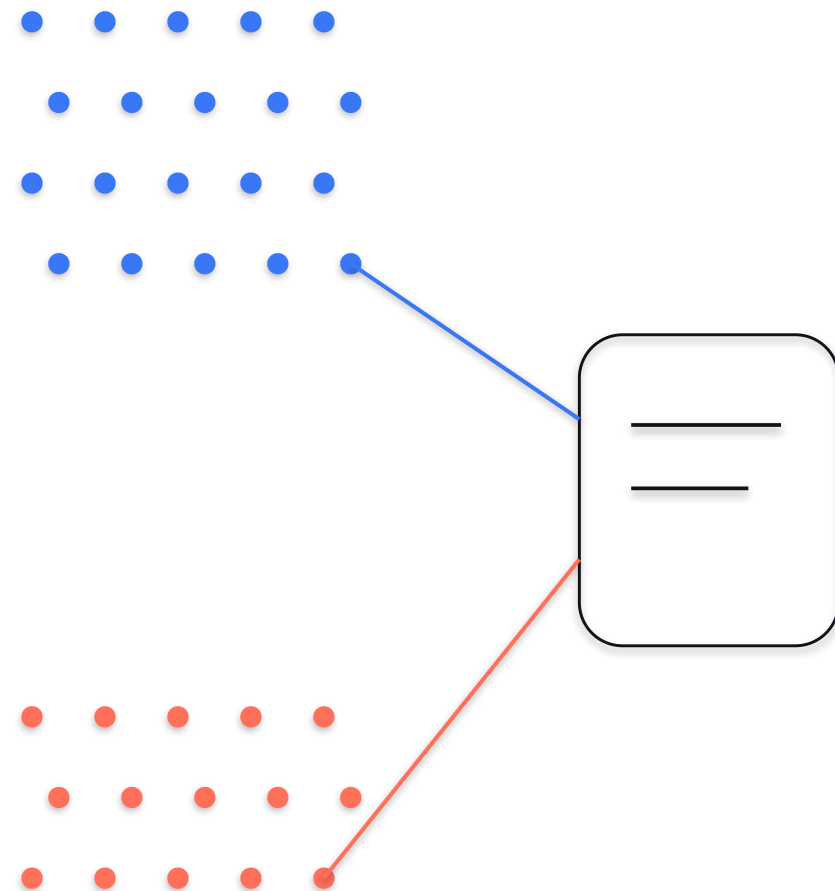


WHO SPEAKS FOR THE CROWD?

Emrecan Ulu / Shi Jingyao



One note. One decision.



- **Sees the note**

One note. One decision.



Elon Musk  
@elonmusk

Subscribe



Have you ever wondered why the Wikimedia Foundation wants so much money?

It certainly isn't needed to operate Wikipedia. You can literally fit a copy of the entire text on your phone!

So, what's the money for? Inquiring minds want to know ...

- Sees the note

- Reads the context

One note. One decision.



- Sees the note



Elon Musk 
@elonmusk

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Readers added context they thought people might want to know

The Wikimedia Foundation is a charitable non-profit, providing free access to Wikipedia. While a text and English-only copy of Wikipedia is about 51GB, including all media and supported languages totals to 428TB. In 2022 Wikimedia had \$154M in revenue with \$145M in expenses.
en.wikipedia.org/wiki/Wikipedia...
wikimediafoundation.org/about/annualre...

- Reads the context



- Rates helpfulness

One note. One decision.



- Sees the note



Elon Musk  
@elonmusk

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en.wikipedia.org/wiki/Wikipedia...
wikimediafoundation.org/about/annualre...



Vivid  
@VividProwess

Subscribe



I am having to constantly run to the nearest bomb shelter with my family, this is not a game, it is now part of our life and reality.

I refuse for our children to grow up living in hell.

Hezbollah will be destroyed.



Readers added context they thought people might want to know

This user is notorious for creating clickbait content and seeking high levels of engagement. User resides in the slums of Mehabullapur, Lucknow, Uttar Pradesh, India and not Israel.

x.com/Free_Those_dar...

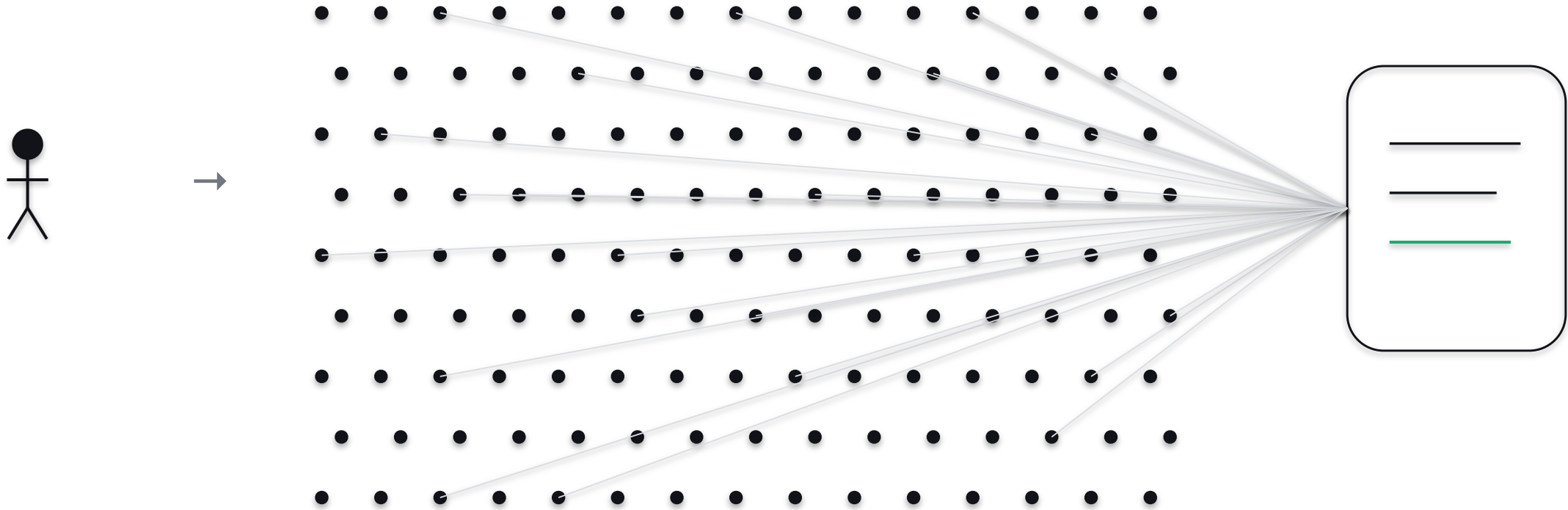
Do you find this helpful?

Rate it

Context is written by people who use X, and appears when rated helpful by others. [Find out more](#)

- Rates helpfulness

Now multiply that by thousands

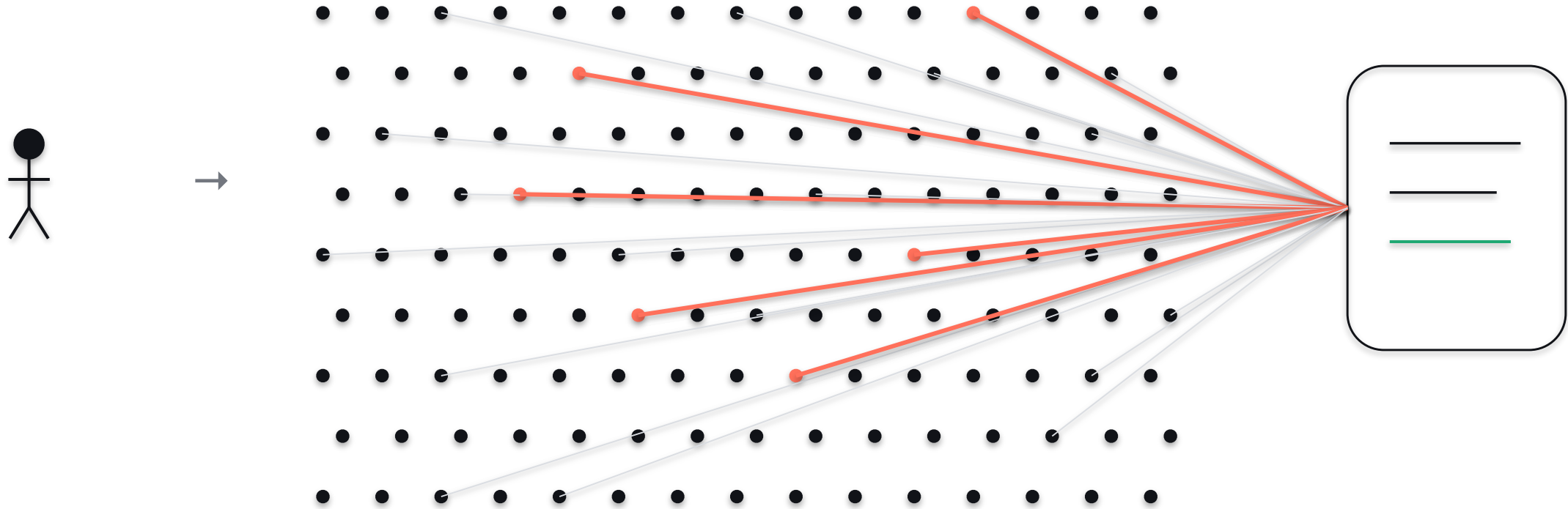


- Same action

- Thousands of raters

- One visibility decision

Now multiply that by thousands



- Same action

- Thousands of raters

- One visibility decision

It reads the rater

AMIGO



It reads the rater

AMIGO

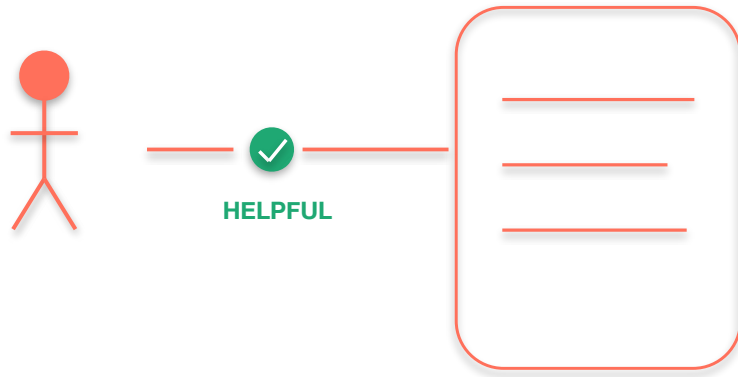


EXPECTED

- Predictable partisan support

It reads the rater

AMIGO



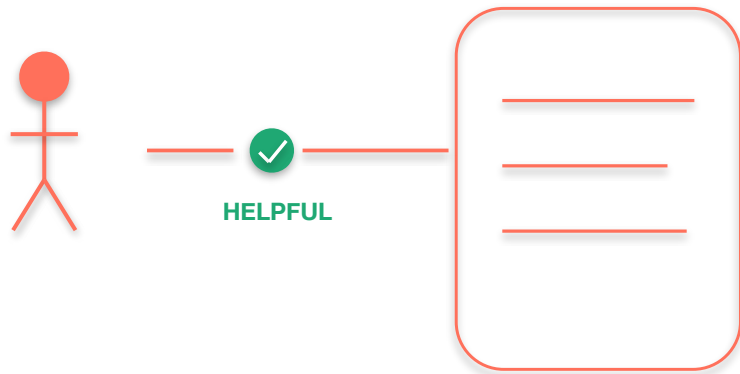
EXPECTED — $f_u \cdot f_n$ — i_n —

- Predictable partisan support

-
- **The same click receives different credit**

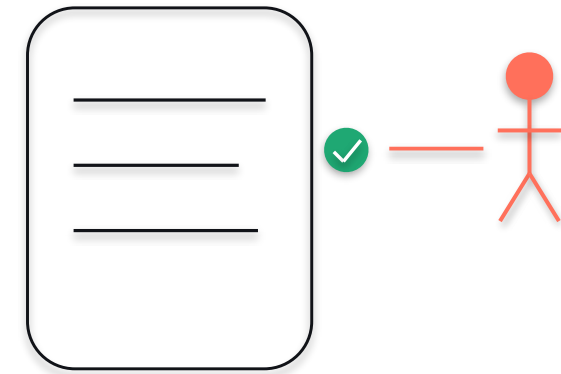
It reads the rater

AMIGO



EXPECTED — $f_u \cdot f_n$ — i_n —

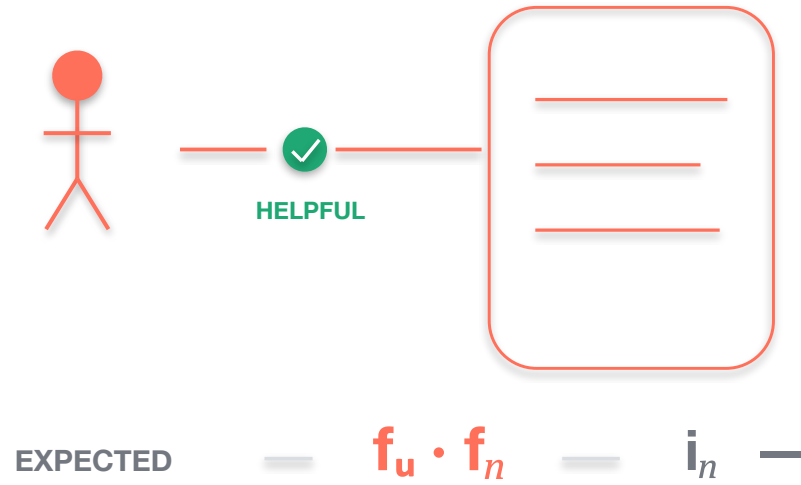
- Predictable partisan support



- **The same click receives different credit**

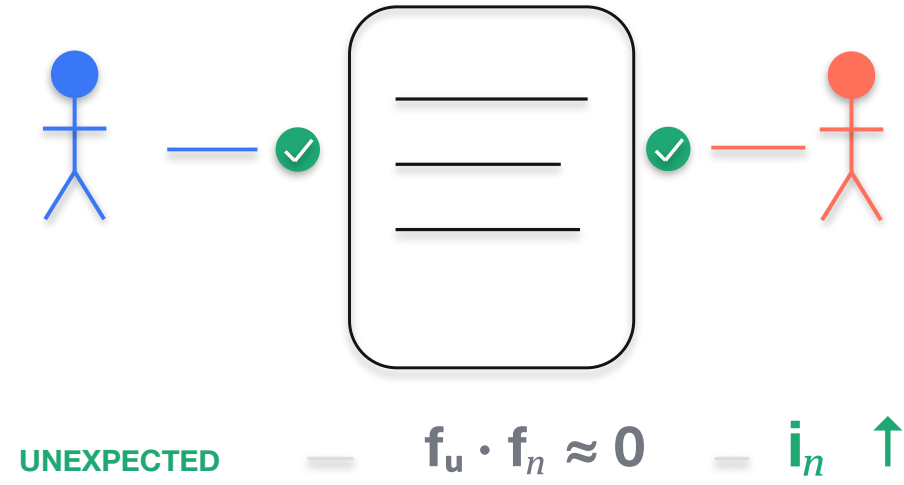
It reads the rater

AMIGO



- Predictable partisan support

ENEMIES SHAKE HANDS



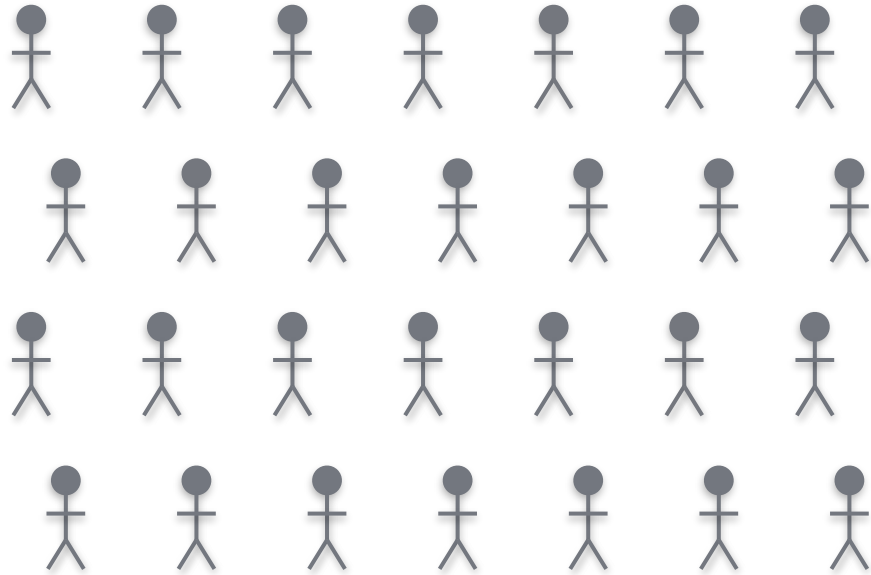
- Cross-viewpoint surprise

- The same click receives different credit

- The model asks who voted

The same hands draw the map

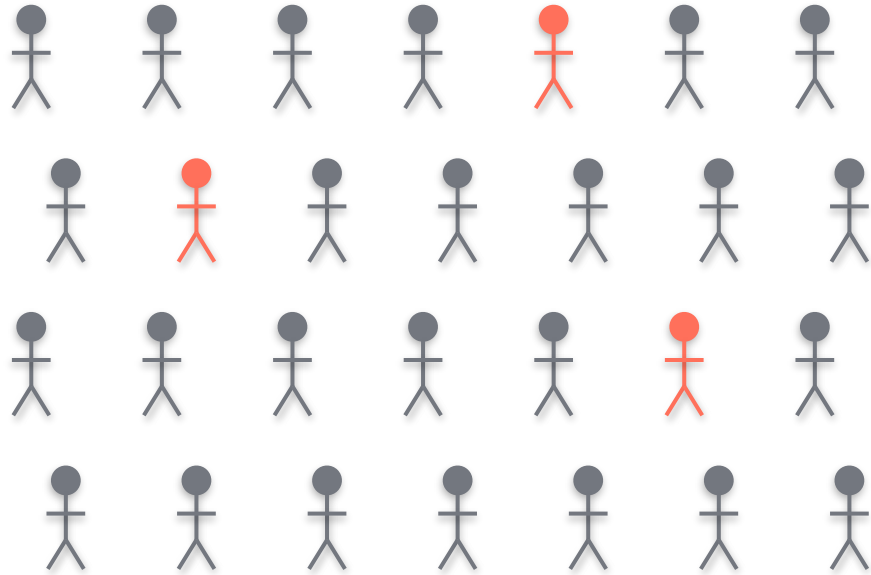
CLASSROOM METAPHOR



- **A few users supply most of the map**
- **The active core becomes ‘the community’**

The same hands draw the map

CLASSROOM METAPHOR

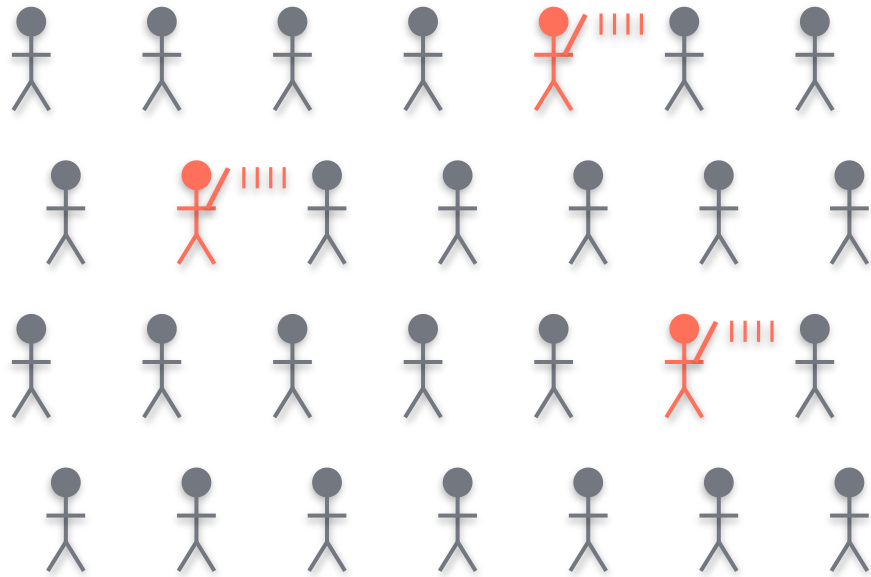


- **A few users supply most of the map**

- **The active core becomes ‘the community’**

The same hands draw the map

CLASSROOM METAPHOR

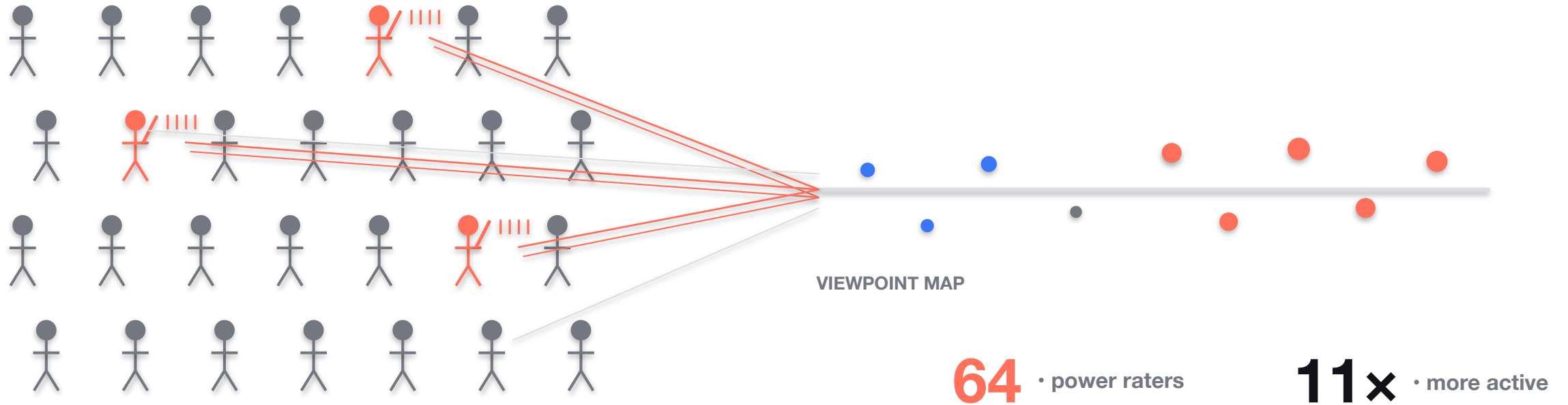


- **A few users supply most of the map**

- **The active core becomes ‘the community’**

The same hands draw the map

CLASSROOM METAPHOR



- A few users supply most of the map
- The active core becomes 'the community'

The handshake is right. The electorate is not.

GOOD INSTINCT



ENEMIES SHAKE HANDS

- Cross-group agreement is valuable

-
- **Keep consent. Make representation explicit.**

The handshake is right. The electorate is not.

GOOD INSTINCT



ENEMIES SHAKE HANDS

- Cross-group agreement is valuable

SELECTION BALANCE

BALANCED



$$\pi A = .50 \quad \pi B = .50$$

$$2(.50)(.50) = 50\%$$

- Keep consent. Make representation explicit.

The handshake is right. The electorate is not.

GOOD INSTINCT



ENEMIES SHAKE HANDS

- Cross-group agreement is valuable

SELECTION BALANCE

BALANCED



$$\pi A = .50 \quad \pi B = .50$$

$$2(.50)(.50) = 50\%$$

OBSERVED



$$\pi A = .90 \quad \pi B = .10$$

$$2(.90)(.10) = 18\%$$

- Keep consent. Make representation explicit.

The handshake is right. The electorate is not.

GOOD INSTINCT



ENEMIES SHAKE HANDS

- Cross-group agreement is valuable

SELECTION BALANCE

BALANCED



$$\pi_A = .50 \quad \pi_B = .50$$

$$2(.50)(.50) = 50\%$$

OBSERVED



$$\pi_A = .90 \quad \pi_B = .10$$

$$2(.90)(.10) = 18\%$$

50% →
18%

THEN IT GETS MESSIER

PARTISAN

$f_u \cdot f_n$ changes the credit

- The same click is interpreted differently

ACTIVITY

π_g follows ratings—not people

- Active users represent each side

- Keep consent. Make representation explicit.

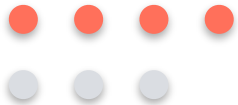
This problem predates platforms

SWITZERLAND

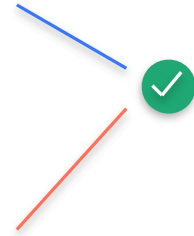
PEOPLE



CANTONS



+



- Double majority

BELGIUM



- Parallel majorities

BOSNIA



- Vital-interest veto

N. IRELAND



- Parallel consent

- No group decides alone

- But online, who are the groups? →

From ratings to constituencies

- Rate notes



●	✓	✓		×		✓	×
●	✓		✓	×		✓	×
●		✓	✓	×	×		×
●	×	×		✓		×	✓
●	×		×	✓	✓		✓
●		×	×	✓	✓	×	
USERS	NOTES						

-
- Similar co-rating patterns → same behavioral constituency

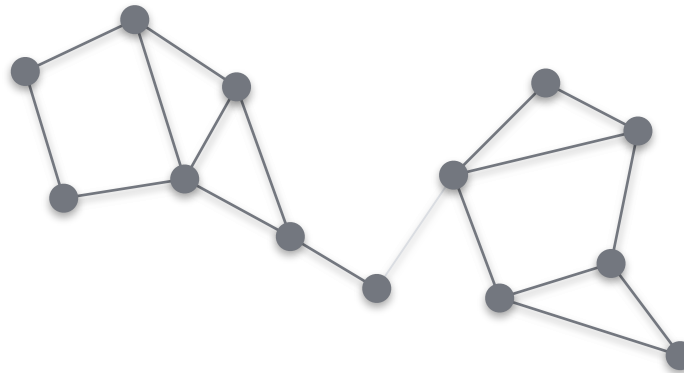
From ratings to constituencies

- Rate notes

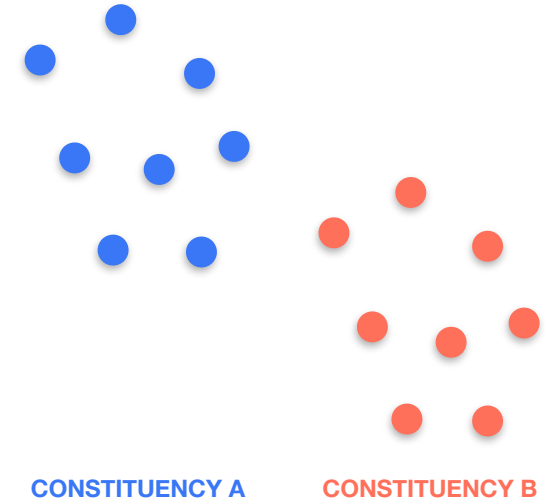


	✓	✓		×		✓	×
	✓		✓	×		✓	×
		✓	✓	×	×		×
	×	×		✓		×	✓
	×		×	✓	✓		✓
		×	×	✓	✓	×	
USERS	NOTES						

- Find matching patterns



- Recover constituencies



• Similar co-rating patterns → same behavioral constituency

Consult constituencies directly



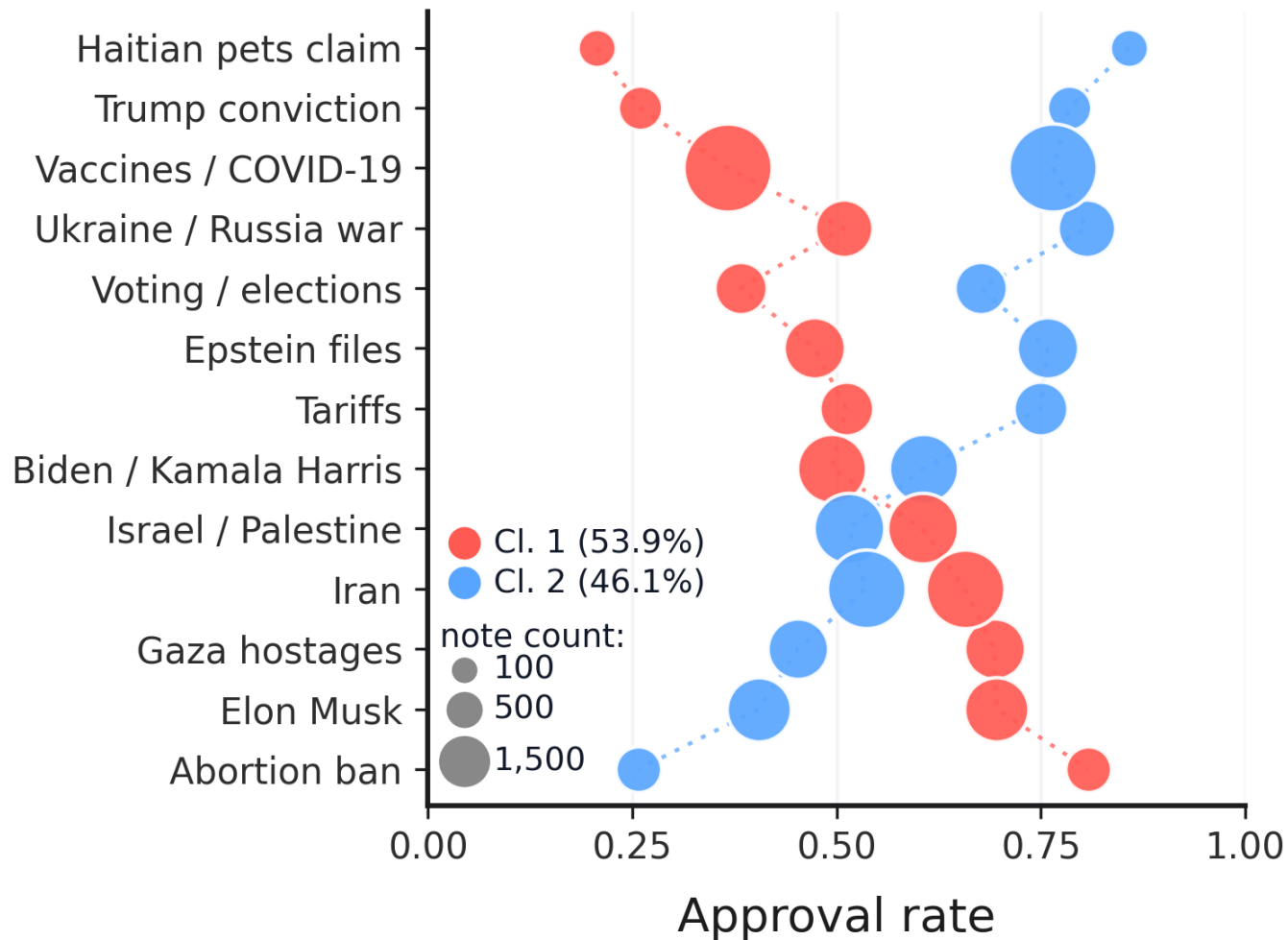
- Recover groups

- Ask each group

- Aggregate explicitly

• Recover groups → consult each one → aggregate explicitly

Disagreement changes by topic



HOW TO READ

- x-position = approval rate
- bubble size = note count
- color = recovered constituency

KEY RESULT

- **Approval leadership flips across topics**
- **Constituencies are content-dependent**
- **The clusters are not simply 'strict' versus 'lenient'**

Four design principles

**P1**

Presence

- Every group enters

**P2**

Non-compensation

- Rejection still matters

**P3**

Symmetry

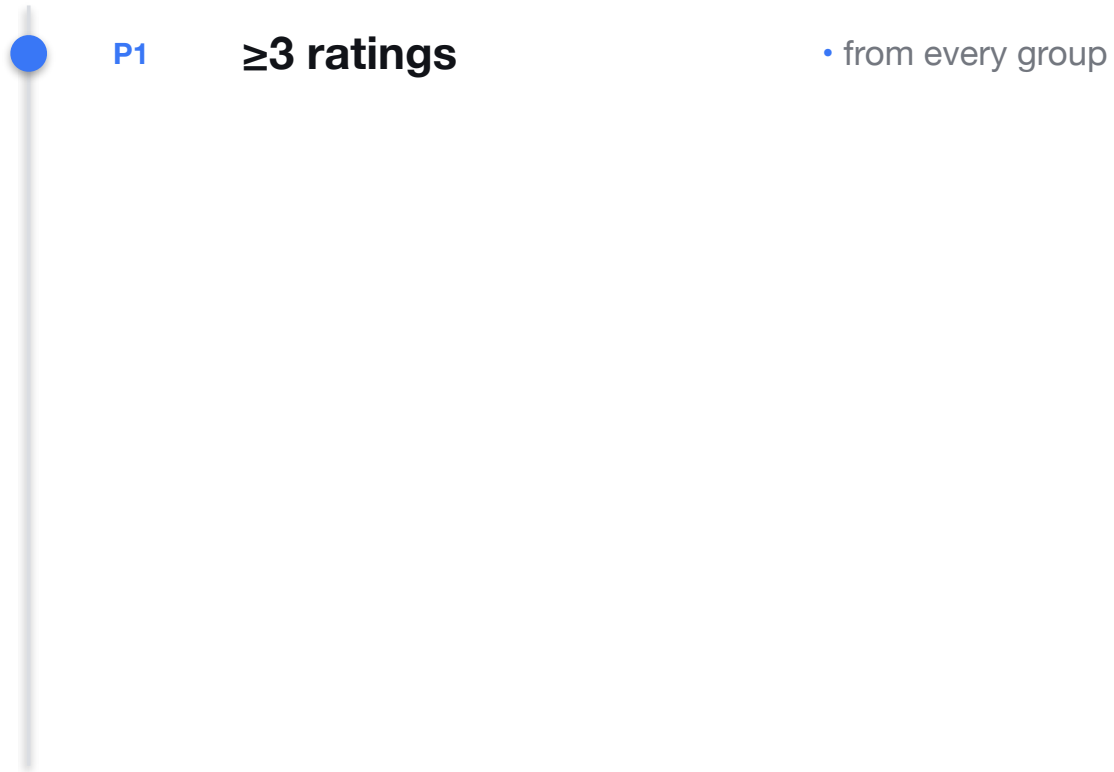
- No default winner

**P4**

Behavioral recovery

- No outside labels

Principles become operations



CHECK 1 · PRESENCE

SAME NOTE

CONSTITUENCY A

90 ratings



CONSTITUENCY B

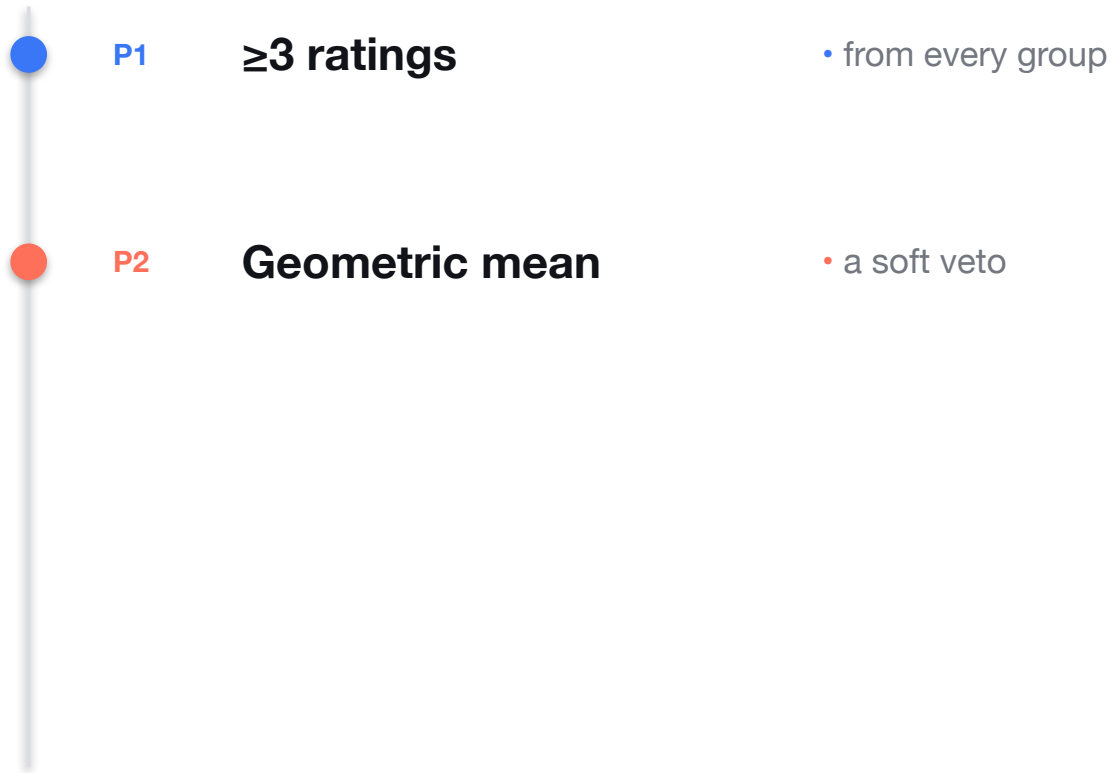
2 ratings



NO SCORE YET

• ≥3 from every group

Principles become operations



CHECK 2 · NON-COMPENSATION

$$A \cdot 81 / 90 = 90\%$$

$$B \cdot 1 / 10 = 10\%$$

POOLED VOTES

$$(81 + 1) / (90 + 10) = 82\%$$

PASS

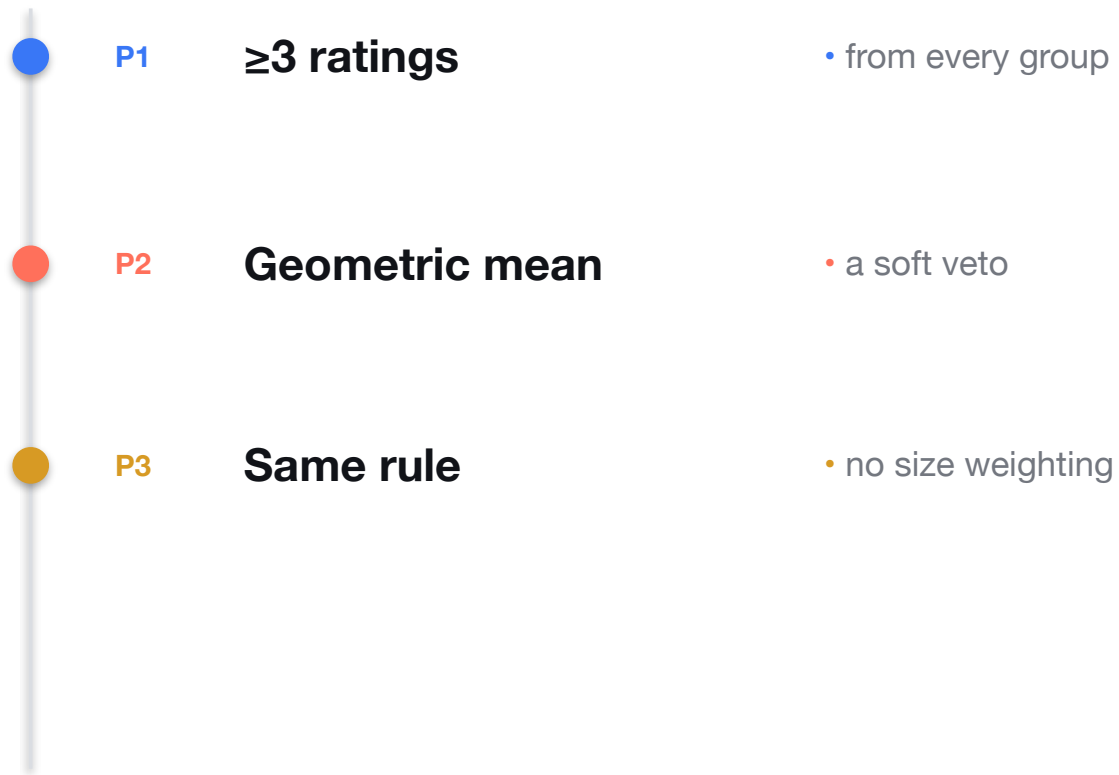
CCA

$$\sqrt{(.90 \times .10)} = 30\%$$

DOES NOT PASS

- Enthusiasm cannot erase rejection

Principles become operations



CHECK 3 · SYMMETRY

90 RATERS

10 RATERS



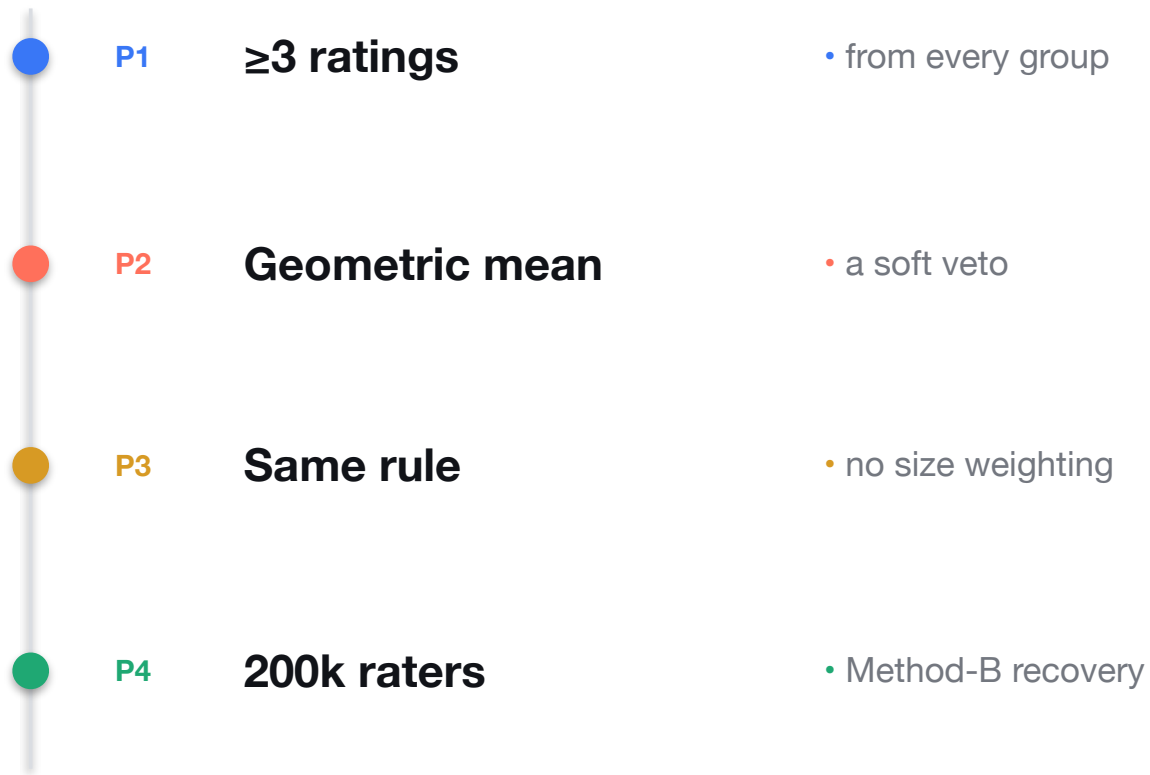
90 raters $\neq$ 9× weight

$$\sqrt{(sA \times sB)} = \sqrt{(sB \times sA)}$$

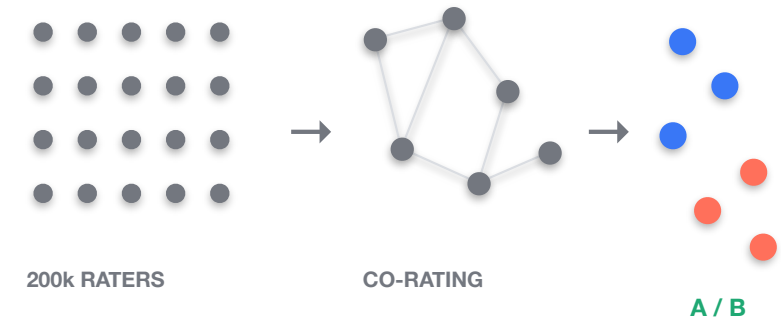
$$\sqrt{(.90 \times .10)} = \sqrt{(.10 \times .90)} = 30\%$$

- Group size does not change the rule

Principles become operations



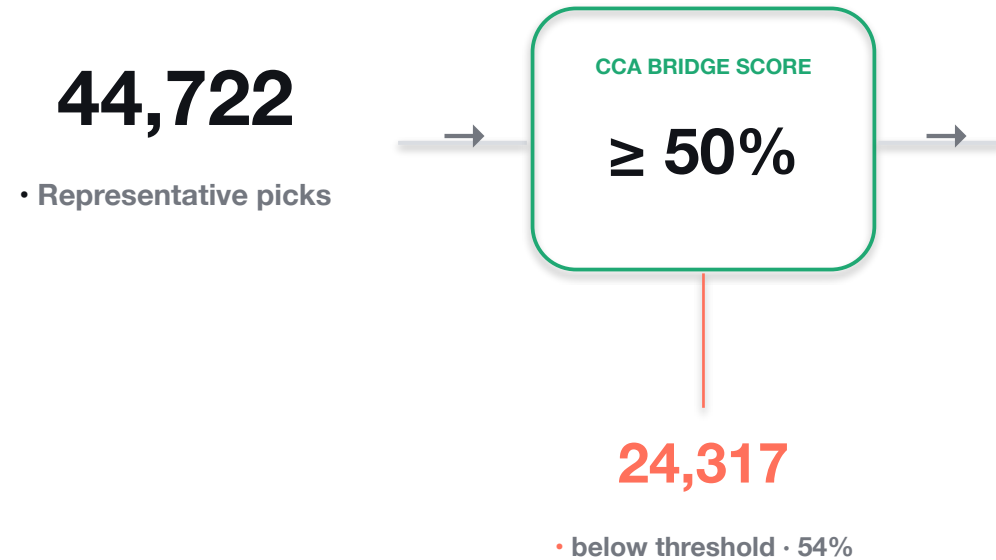
CHECK 4 · BEHAVIORAL RECOVERY



METHOD B

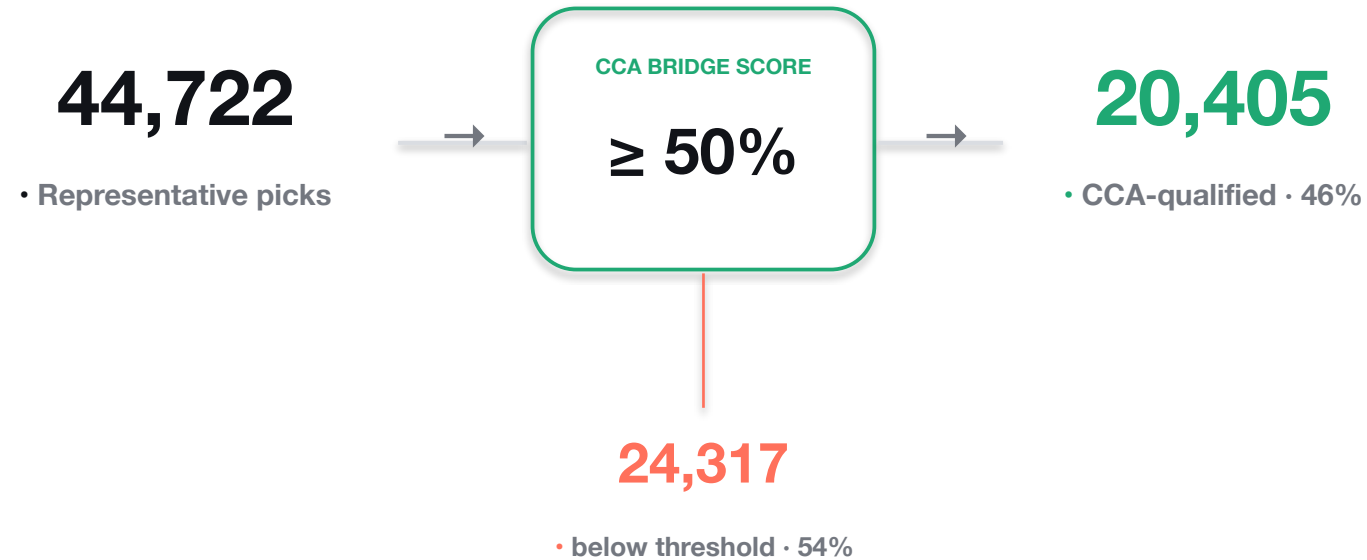
- No party · country · ideology labels

What the platform shows—and hides



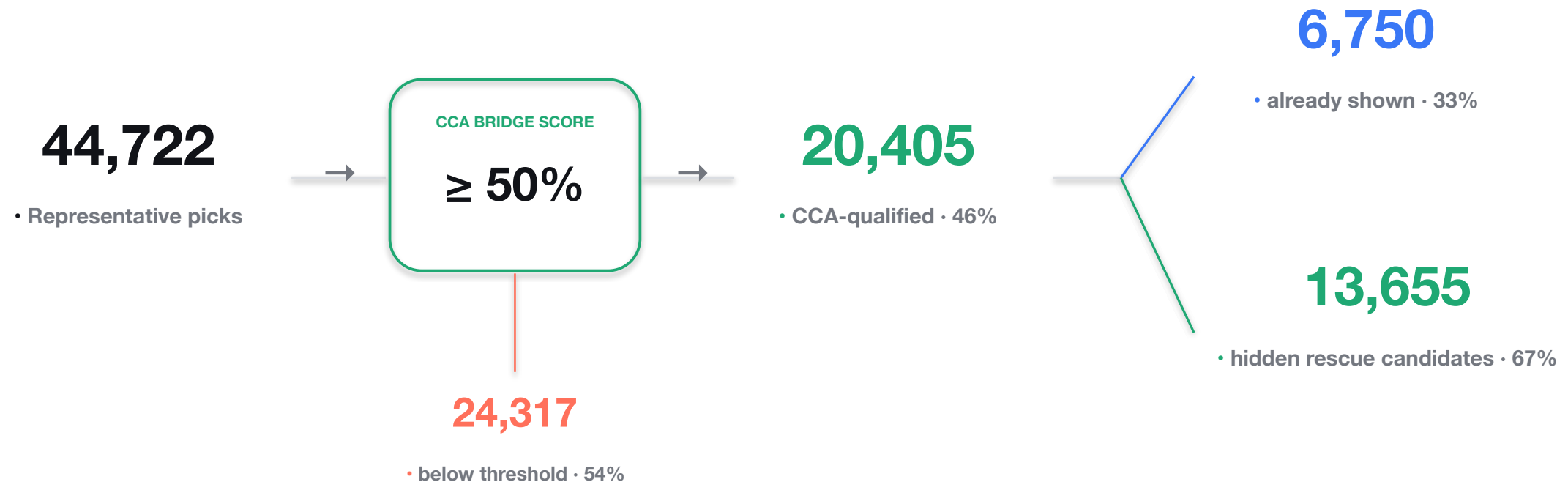
-
- CCA identifies one qualified pool—some already shown, others hidden

What the platform shows—and hides



-
- CCA identifies one qualified pool—some already shown, others hidden

What the platform shows—and hides



- CCA identifies one qualified pool—some already shown, others hidden

Gemma Validation

STAGE 1

Nature + sourcing

PASS • sourced factual information

RECHECK • opinion / speculation

STAGE 1.5 • SOURCED FACTUAL CORE

STOP • unsourced context / claim

• hostile / derogatory

• irrelevant / spam

VISIBLE EXTERNAL SOURCE POINTER REQUIRED

STAGE 2

Rescue worthiness

0–100

• source traceability

• claim–source connection

• clarity and neutrality

• constructive presentation

≥50 → **VALIDATED**

- **Missing sourcing is a validation failure—not proof that the claim is false**

A 30.7B model, pinned end to end

30.7B

DENSE PARAMETERS · GEMMA 4 31B IT

2× L40S

46,068 MiB EACH · TENSOR PARALLEL 2

64

CONCURRENT REQUESTS · MAX SEQUENCES

15h 51m

FIRST OUTPUT → FINAL OUTPUT

MODEL + DECODING

- google/gemma-4-31B-it
- revision 518276fb130dc81caf9a4f772e65e63ef2526493
- BF16 · no quantization
- thinking on · gemma4 reasoning parser
- temperature 0.2
- seed = SHA-256(stage, note, attempt)
- 8,192 context · 4,096 completion

SCCKN + SERVING

- AMD EPYC host · 8 CPU slots
- 64 GB h_vmem / slot · ~52 GB observed peak
- CUDA 13.2 · GPU utilization target 90%
- vLLM 0.25.0 · async scheduling
- PyTorch 2.11.0+cu130
- Transformers 5.13.1
- 32,768 batched tokens · prefix cache on
- language-model-only · local OpenAI-compatible endpoint

INFERENCE CONTRACT + AUDIT

- one note → one independent user prompt
- no system prompt · no guided generation
- strict JSON validation after generation
- no tweet context · no URL fetching
- no retrieval · embeddings · vector store
- no Gabriel scaffold or label in context
- ≤3 attempts · durable per-attempt records
- 25,734 judgments · 25,804 total attempts
- 32.1M tokens · 99.75% first-attempt valid

ACTIVE PHASES 7h 18m STAGE 1 + 36m STAGE 1.5 + 4h 35m STAGE 2 = 12h 28m OBSERVED ACTIVE TIME LOCAL vLLM · NO HOSTED API

From candidates to validated rescues

13,655



GEMMA 4 31B IT

COMPLETED CANONICAL RUN

• 8,527 strict route

• 31 recall route

10,376 / 10,376 complete · zero unresolved Stage 2 judgments

From candidates to validated rescues

13,655

10,376

• CCA candidates

frozen universe

3,279 CONTENT STOP

- 1,423 opinion / factual core absent
- 1,340 irrelevant / trivial / spam
- 373 unsourced context / claim
- 142 hostile / derogatory
- 1 unresolved

• Stage 2 judgments

10,096 strict + 280 recall

1,818 BELOW 50

- 90–100 **outstanding** · clear, traceable, self-contained
- 70–89 **strong** · well-supported and useful
- 40–69 **mixed** · useful elements, material gaps
- 10–39 **weak** · limited support or clarity
- 0–9 **minimal** · little usable justification
- ≥50 **validated** · display-quality threshold

GEMMA 4 31B IT

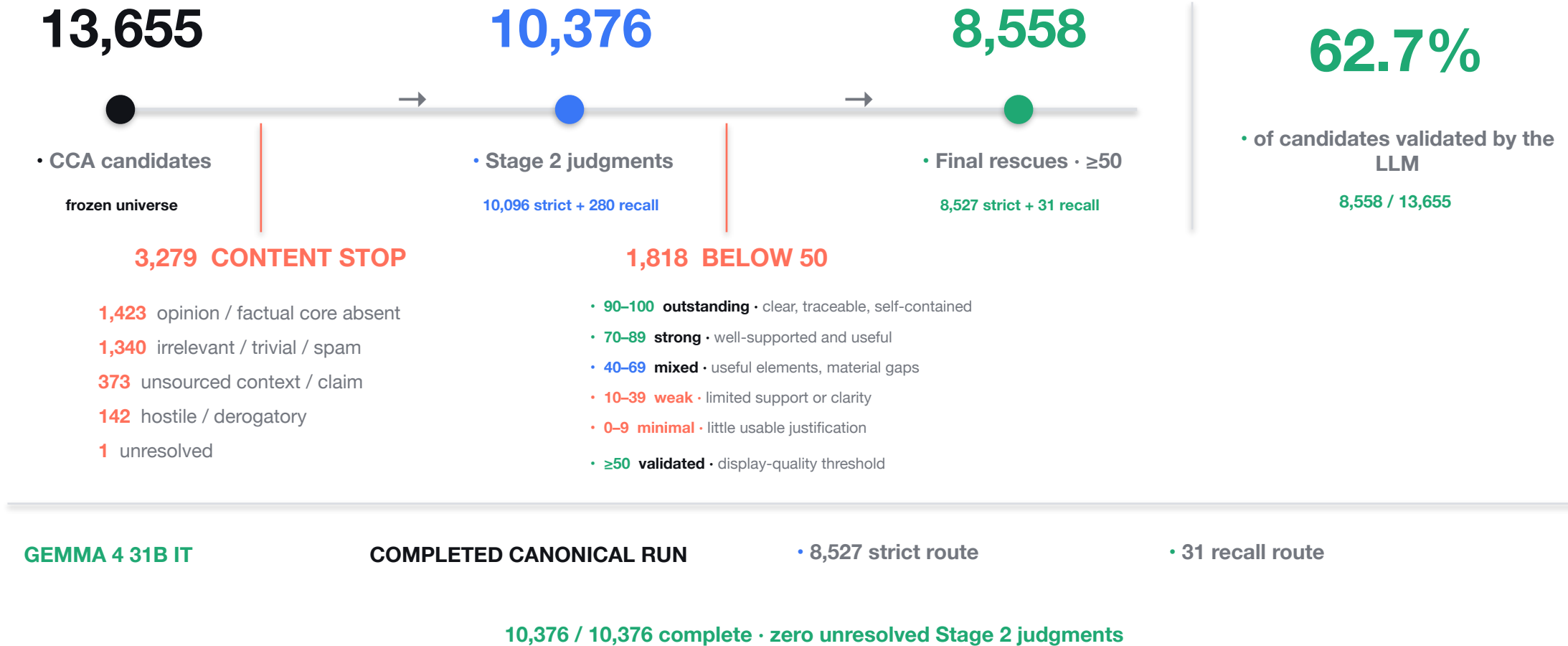
COMPLETED CANONICAL RUN

• 8,527 strict route

• 31 recall route

10,376 / 10,376 complete · zero unresolved Stage 2 judgments

From candidates to validated rescues



High approval. Wrong battle.

PLATFORM-SHOWN NOTE

NOTE 1988138343385731249

CURRENTLY_RATED_HELPFUL

92.2% overall approval

C0 98.3%

C1 60.9%

282 ratings

$\sqrt{(.983 \times .609)} = 77.4\%$

PLATFORM
SHOWS

The Battle of Mu'tah was fought between the Byzantine Empire and the First Islamic State. The true size of the armies is considered to be around 10,000 for the Byzantines and 3000 for the Muslims. It ended with a Byzantine victory.

SOURCES [Wikipedia \(Mu'tah\)](#) · [Islam365](#)

• Platform approval favored the wrong battle; CCA selected the relevant correction

High approval. Wrong battle.

PLATFORM-SHOWN NOTE

NOTE 1988138343385731249

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SOURCES Wikipedia (Mu'tah) · Islam365

CCA / REPRESENTATIVE PICK

NOTE 1988182086097006685

NEEDS_MORE_RATINGS

91.2% overall approval

C0 95.2% C1 73.7%

204 ratings

$\sqrt{(.952 \times .737)} = 83.7\%$

CCA PICKS

The post refers to the Battle of Yarmouk (636 AD), but reliable estimates put Muslim forces at 15,000–40,000, not 3,000, versus 15,000–150,000 Byzantines. 3,000 Muslims fought at the earlier Battle of Mu'tah (629 AD).

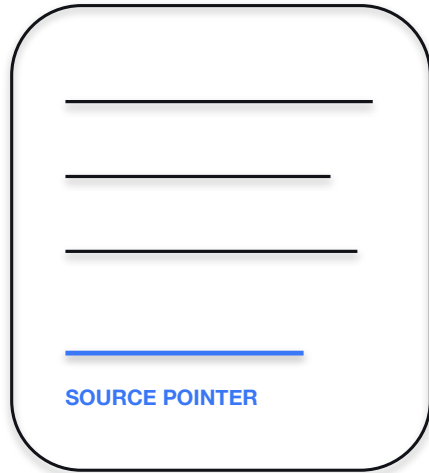
SOURCES Wikipedia (Yarmouk, Mu'tah) · Britannica

• Platform approval favored the wrong battle; CCA selected the relevant correction

Thank You

What makes a note worth rescuing?

MODEL SEES

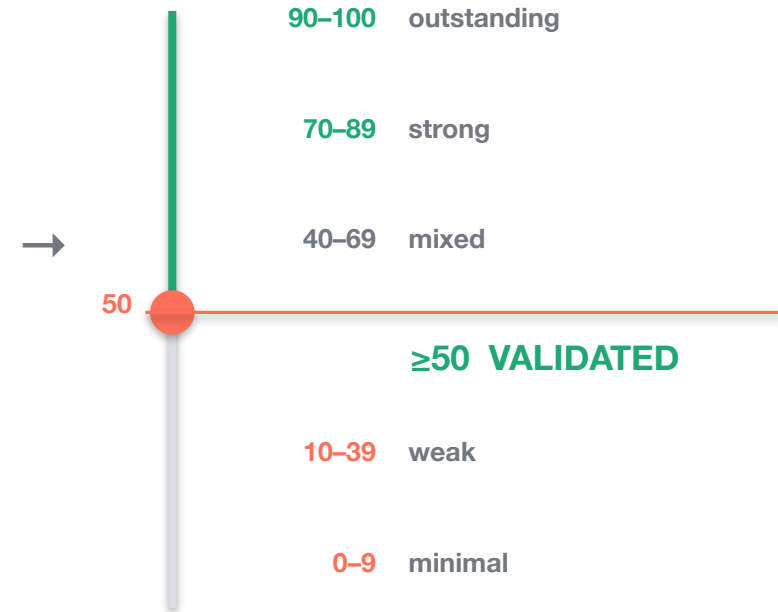


- note text
- visible source pointer

HOLISTIC JUDGMENT

- specific + traceable source pointer
- clear claim ↔ source connection
- self-contained explanation
- factual + neutral wording
- concise + constructive presentation

0-100 SCORE



- No URL opening · No original post · Visible note quality only